SageMaker Porting Debugging Harvest (Updated README Addendum)

This document harvests commands, event sequences, failure signatures, fixes, and operational guidance derived from the chat, and updates the original README with practical lessons learned.

1. Critical Lessons Learned

```
1) Keep environments separate:
- ML training venv: numpy/pandas/scikit-learn/xgboost/joblib only
- AWS CLI: install globally (brew install awscli)
- Optional AWS SDK venv: boto3 + sagemaker only

2) For SageMaker Script Mode, model.tar.gz must contain:
./model.pkl
./preprocessor.joblib
./label_map.json
./code/inference.py
./code/requirements.txt

3) Common failure sequence encountered:
- Missing /opt/ml/model/code -> packaging error
- numpy.core.multiarray failed to import -> dependency conflict
- No module named '_loss' -> sklearn training/serving version mismatch

4) Endpoint creation may take 10–25 minutes to fail because SageMaker retries unhealthy containers.
```

2. Packaging Commands (Preferred)

```
cd dist/sagemaker

rm -rf /tmp/benefits_sm_pkg
mkdir -p /tmp/benefits_sm_pkg/code

cp model/* /tmp/benefits_sm_pkg/
cp code/*.py /tmp/benefits_sm_pkg/code/
cp code/requirements.txt /tmp/benefits_sm_pkg/code/

tar -czf model.tar.gz -C /tmp/benefits_sm_pkg .

tar -tzf model.tar.gz

aws s3 cp model.tar.gz \
 s3://xt-benefits-model-bucket-fjg/benefits-classifier/v1/model.tar.gz \
 --region us-east-2
```

3. Dependency Pins Found Safer

```
requirements.txt (recommended)
xgboost==1.7.6
joblib==1.3.2

If needed for stricter pinning:
numpy==1.23.5
pandas==1.5.3
scikit-learn==1.2.1
```

4. Recreate SageMaker Model

```
ROLE_ARN="arn:aws:iam::219217499832:role/SageMakerExecutionRole"
REGION="us-east-2"
MODEL_URI="s3://xt-benefits-model-bucket-fjg/benefits-classifier/v1/model.tar.gz"
IMAGE="257758044811.dkr.ecr.us-east-2.amazonaws.com/sagemaker-scikit-learn:1.2-1-cpu-py3"

aws sagemaker create-model \
 --model-name benefits-eligibility-v1 \
 --execution-role-arn "${ROLE_ARN}" \
 --primary-container "Image=${IMAGE},ModelDataUrl=${MODEL_URI},Environment={SAGEMAKER_SUBMIT_DIRECTORY=/opt/ml/
 --region "${REGION}"
```

5. Endpoint Create/Delete Commands

```
Create endpoint config:
aws sagemaker create-endpoint-config \
 --endpoint-config-name benefits-eligibility-config \
 --production-variants VariantName=AllTraffic,ModelName=benefits-eligibility-v1,InitialInstanceCount=1,Instance
 --region us-east-2

Create endpoint:
aws sagemaker create-endpoint \
 --endpoint-name benefits-eligibility-endpoint \
 --endpoint-config-name benefits-eligibility-config \
 --region us-east-2

Delete endpoint:
aws sagemaker delete-endpoint \
 --endpoint-name benefits-eligibility-endpoint \
 --region us-east-2

Delete endpoint config:
aws sagemaker delete-endpoint-config \
 --endpoint-config-name benefits-eligibility-config \
 --region us-east-2

Delete model (metadata only; does not remove S3 artifact):
aws sagemaker delete-model \
 --model-name benefits-eligibility-v1 \
 --region us-east-2
```

6. Monitoring / Diagnostics Commands

```
Status:
aws sagemaker describe-endpoint \
 --endpoint-name benefits-eligibility-endpoint \
 --region us-east-2 \
 --query '{Status:EndpointStatus,FailureReason:FailureReason}'

Poll:
while true; do
 aws sagemaker describe-endpoint \
 --endpoint-name benefits-eligibility-endpoint \
 --region us-east-2 \
 --query '{Status:EndpointStatus,FailureReason:FailureReason}'
 sleep 30
done

Newest log stream:
```

```
aws logs describe-log-streams \
 --log-group-name "/aws/sagemaker/Endpoints/benefits-eligibility-endpoint" \
 --region us-east-2 \
 --order-by LastEventTime \
 --descending \
 --max-items 1
```

7. Local inference.py Validation

```
python - <<'PY'
import sys
sys.path.insert(0,"code")
import inference
print("import OK")
model=inference.model_fn("model")
print("model load OK")
PY
```

8. Invoke Endpoint (AWS CLI v2 Fix)

```
Use --cli-binary-format raw-in-base64-out

aws sagemaker-runtime invoke-endpoint \
 --endpoint-name benefits-eligibility-endpoint \
 --content-type application/json \
 --body '{"instances":[{"age":34,"employment_type":"full_time"}]}' \
 --cli-binary-format raw-in-base64-out \
 output.json \
 --region us-east-2
```

9. README Corrections to Original

```
Update original README:
- Remove 'pip install awscli boto3 sagemaker' from ML venv instructions.
- Add warning not to install awscli inside model-training venv.
- Add troubleshooting section covering:
  a) /opt/ml/model/code missing
  b) numpy.core.multiarray failed to import
  c) sklearn _loss pickle mismatch
- Add note that local success can still fail in SageMaker due to container version mismatch.
- Add recommendation to match training sklearn version to serving container.
```

10. Recommended Next Validation After InService

```
After endpoint reaches InService:
1) Invoke with test payload
2) Compare prediction to local model
3) Wrap with FastAPI or API Gateway if exposing as service
4) Tag endpoint and document endpoint name for consumers
```